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COLLEGE OF ENGINEERING
Akurdi, Pune

Presentation
On
Subject: UI and UX Design
by

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Unit 1: Introduction to UI and UX

Sessions

User Research, Personas, and Journey Mapping: User research techniques: Interviews, surveys, contextual inquiry, creating empathy maps and personas, Task flows and user journey mapping, identifying pain points and defining opportunities.

Learning Outcomes of the session:

After completing this unit, students will be able to:

- **Understand and apply user research techniques (interviews, surveys, contextual inquiry)**
- **Create empathy maps and user personas based on research insights**
- **Design task flows and user journey maps**
- **Identify user pain points and UX issues**
- **Define UX improvement opportunities using user-centered thinking**

Content:

Content	Learning / Methodology	Faculty Approach	Typical Student Activity	Skill/ Competency Developed
User Research, Personas, and Journey Mapping: User research techniques: Interviews,surveys,contextual inquiry, creating empathy maps and personas, Task flows and user journey mapping, identifying pain points and defining opportunities.	PPT, Experimentation	Explain	Understand	Critical Thinking - Observation

UI & UX DESIGN

Syllabus



What is UI/UX?

UI → User Interface



You know exactly where to press or click without even thinking about it.

UX → User Experience



Does it make your life easier or harder?



UX



UX





UX RESEARCH

Where to start...?

What are the best research tools?

...It depends:

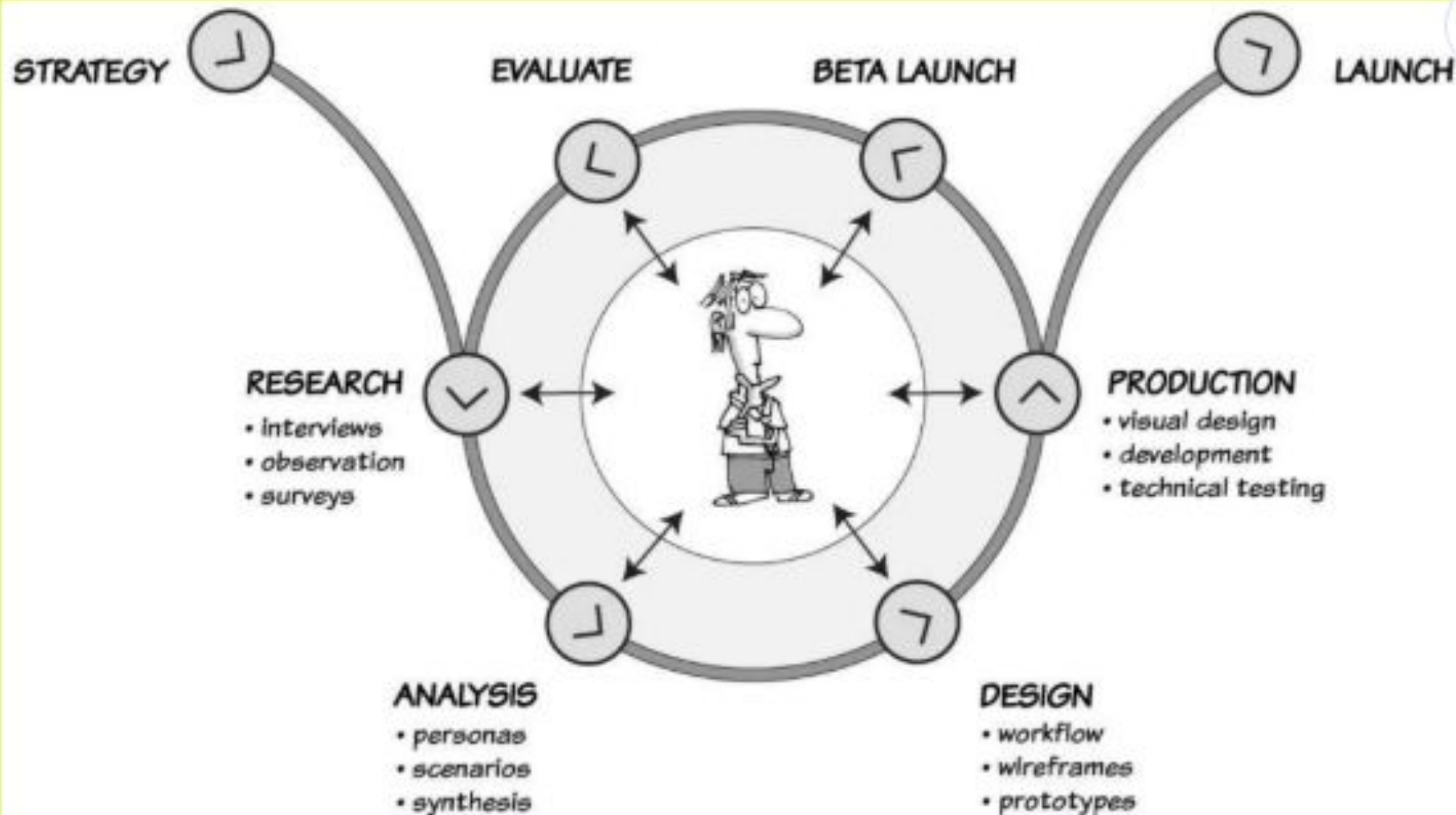


ON **WHAT** YOU NEED TO TEST

=

WHAT IS THE **STAGE** OF YOUR PROJECT





There are many research methods useful in the process of a digital product development

Different methods are applicable in **different phases** of a project

**TEST EARLY. TEST OFTEN.
TEST WITH REAL USERS.
ITERATE.**

User research

User research is the methodic study of target users—including their needs and pain points—so designers have the sharpest possible insights to make the best designs. User researchers use various methods to expose problems and design opportunities and find crucial information to use in their design process.

User research is a crucial part of the design process. It helps to bridge the gap between what we think users need and what users actually need.

Two Approaches to User Research

1. Attitudinal

Listen to what users say.



2. Behavioral

Observe what users do.





User research techniques:

A UX research techniques is a way of generating insights about your users, their behavior, motivations, and needs.

User Research Techniques are ways to **understand users** –

what they need, what problems they face, and how they use a product or system.

In simple words:

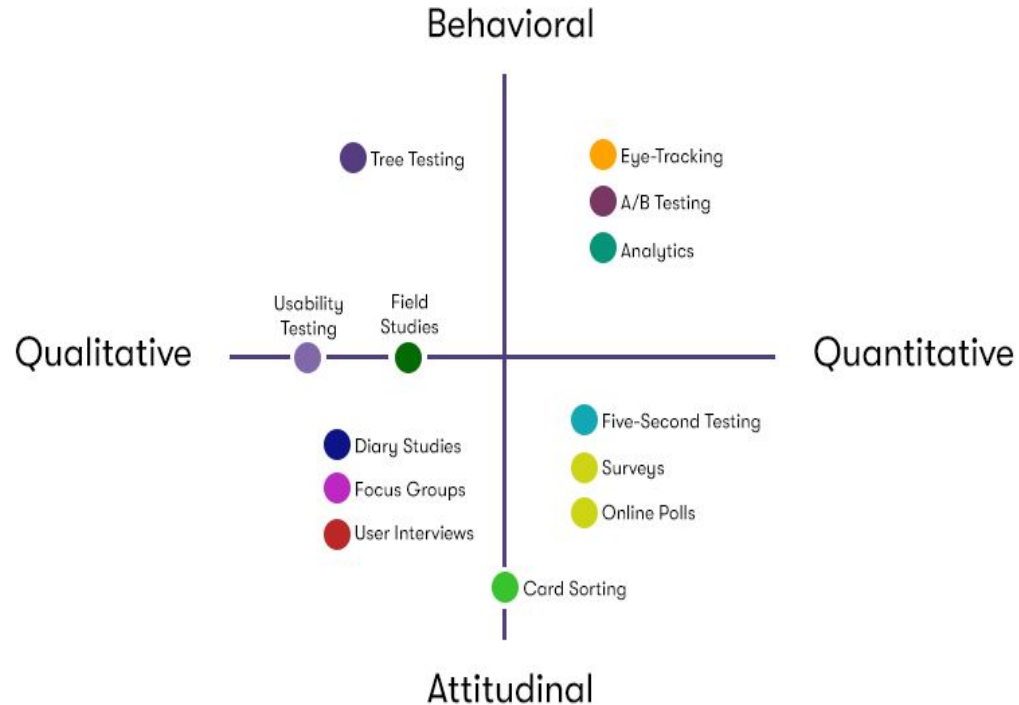
User research helps us design systems that are useful, easy, and user-friendly.

Types of User Research Techniques

There are mainly two types:

1. Qualitative – Based on words and observations (Why, How)
2. Quantitative – Based on numbers and data (How many, How often)

UX Research Methods



Types of User Research Techniques

There are mainly **two types**:

1. **Qualitative** – Based on words and observations (Why, How)
2. **Quantitative** – Based on numbers and data (How many, How often)

1. Qualitative Techniques

These help us understand **feelings, opinions, and problems**.

1. Interviews

Talking directly with users.

Example:

Ask 5 students:

- “What problem do you face while using Moodle?”

3. Focus Group

Group discussion with users.

Example:

10 students discuss:

- What features they want in college app.

4. Usability Testing

Give tasks and see how users perform.

Example:

Task: “Find your result on website.”

You observe how long it takes and where they click.

5. Diary Study

Users write their experience daily.

Example:

Students write problems they face daily while using online library portal.

2. Quantitative Techniques

These use numbers and data.

1. Surveys / Questionnaires

Asking many users using forms.

Example:

Google Form to 100 students:

Rate website from 1 to 5.

2. Analytics

Check system data.

Example: Website shows:

70% users leave on login page.

RESEARCH TECHNIQUES

3. A/B Testing

Compare two designs.

Example:

Version A: Blue button

Version B: Green button

See which one gets more clicks.

4. Heatmaps

Shows where users click most.

Example:

Heatmap shows users never click “Help” button.

Simple College Website Example

Problem:

Students say college website is confusing.

Step 1 – Interview

Talk to 10 students → they say “Notices are hard to find”.

Step 2 – Survey

Survey 100 students → 80% say same problem.

Step 3 – Usability Test

Ask users to find notice → most fail.

Step 4 – Redesign

Make “Notices” button bigger.

Step 5 – A/B Test

Check old vs new design → new one works better.

Qualitative Research

Ethnographic [field studies](#) and interviews are examples of methods that can help you build a deep understanding of **why** users behave the way they do (e.g., why they leave a website so quickly).

For instance, you can interview a small number of users and get sharp insights into their shopping habits by asking them open-ended questions. [Usability testing](#) is another dimension of this type of research (e.g., examining users' stress levels when they use a certain design).

[Qualitative research](#) requires great care. As it involves collecting non-numerical data (e.g., opinions), your own opinions might influence findings.

Quantitative Research

With more structured methods such as [surveys](#), you gather measurable data about **what** users do and [test assumptions](#) you developed from qualitative research.

An example is to use an online survey to ask users questions about their shopping habits (e.g., “Approximately how many items of clothing do you buy online per year?”). You can use this data to find patterns within a large user group. In fact, the larger the sample of representative test users is, the more likely you’ll have a statistically reliable way of assessing the target user population.

Regardless of the method, with careful research, you can gather objective and unbiased data. Nevertheless, quantitative data alone *cannot* expose deeper human insights.

Attitudinal vs Behavioral Research

We can also split user research into two approaches:

1. **Attitudinal** – you listen to users' words (e.g., in interviews).
2. **Behavioral** – you watch their actions through observational studies.

Two Approaches to User Research

1. Attitudinal
Listen to what
users say.



2. Behavioral
Observe what
users do.

Research methods vary from **explorative** techniques (in early phase) to **validation** techniques (in late phases) based on research objectives set up according to specific **PRODUCT DEVELOPMENT STAGE**



FROM
EXPLORATION

TO VALIDATION

Exploring and gathering users':

- Needs and motivations
- Expectations and desires
- User stories
- Users' painful moments and the „moments of true“
- Users' demographics, U&A analysis
- Market and sector insights
- Competitors analysis and best practices, etc.



Validating and testing:

- Product value and concept idea
- IA / taxonomy testing
- Interaction flow testing
- Overall usability testing on:
- Paper or clickable prototypes
- Visual (overall) design evaluation
- Product potential testing, etc.

8 COMMON METHODS USED IN UX RESEARCH.

1. Creating personas
2. User journey mapping
3. 2-minutes interview
4. Card sorting
5. Competitive audit
6. Heuristic evaluation
7. Usability testing: Prototype testing
8. Heat Maps

CREATING PERSONAS



Before starting any product development it is crucial to answer: **who are the people we are designing the solution for?**



Personas support **user-centered design** throughout a project's lifecycle by making characteristics of key user segments more salient



WHAT IS PERSONA?



A fictional character that **represents the group of users** described in a such way that we can **picture** him or her as an **alive tangible person**

PURPOSE

To understand better **who the users are** in order to fulfill their expectations and needs via newly developed product



Name of persona.....

Demography

- Age
- Family status
- Education
- Occupation and income
- Where he lives

Needs and motivators

- What he aspires for, wishes, dreams
- What kinds of problems he solves
- What are his personal and work goals
- What motivates him
- Who is an idol figure for him
- What is important for him at work
- What frustrates him

Credo / quote

Appearance and personality

- His characteristic traits
- How he communicates
- What is important to him
- What he likes/dislikes
- What kind of problem he solves

Interests

- What he does for fun
- Where you can meet him
- What he reads and watch
- Tools he uses (notebook, i-watch, i-pad...)
- Applications and web sites he fancies
- Social nets he uses

Clark Andrews

AGE 26

OCCUPATION Software Developer

STATUS Single

LOCATION San Jose, CA

TIER Experiment Hacker

ARCHETYPE The Computer Nerd

Friendly

Clever

Go-Getter

Motivations

Incentive

Fear

Achievement

Growth

Power

Social

Goals

- To cut down on unhealthy eating and drinking habits
- To measure multiple aspects of life more scientifically
- To set goals and see and make positive impacts on his life

Frustrations

- Unfamiliar with wearable technology
- Saturated tracking market
- Manual tracking is too time consuming

Bio

Aaron is a systems software developer, a "data junkie" and for the past couple years, has been very interested in tracking aspects of his health and performance. Aaron wants to track his mood, happiness, sleep quality and how his eating and exercise habits affects his well being. Although he only drinks occasionally with friends on the weekend, he would like to cut down on alcohol intake.

Personality

Extrovert

Introvert

Sensing

Intuition

Thinking

Feeling

Judging

Perceiving

Technology

IT & Internet

Software

Mobile Apps

Social Networks

Brands



"I feel like there's a smarter way for me to transition into a healthier lifestyle."

PROCESS TO CREATE PERSONAS

1. Organize the **team cross-functional workshop**: using the team knowledge to **make a raw draft** of personas
2. Do **research**: interview (possibly combine with **observations**) 2 – 6 people per persona to **verify** and **tune** the team assumptions and knowledge about personas
3. Have **presentation**: show the **final image of users' personas** to the whole team, so each project member has the *consistent knowledge* of the users target groups usable during *the whole project cycle*

What is it?

Technique used to define and describe your groups of potential or current customers / users

When to do it?

In the very beginning of the project

RECAP

Why to use it?

To ensure that all project members know who the target user is throughout the whole project cycle

Objectives?

To understand better your users' needs and expectations

USER JOURNEY MAPPING





A user journey map is a **visualization** of the **process** that a person goes through in order to accomplish a **goal**.

It's used for understanding and addressing customer **needs** and **pain points**

Users journey map consists of:

1. PERSONA: THE USER / THE CUSTOMER
2. USERS' STORIES / SCENARIOS
3. TOUCHPOINTS AND DEVICES CUSTOMERS USE
4. WHAT THE USERS THINK
5. WHAT THE USERS FEEL
6. INSIGHTS + OPPORTUNITIES



OPTIMAL PROCESS TO CREATE USERS' JOURNEY



1. Organize a **team cross-functional workshop** in order to:
 - a. Define a **use case** you want to explore - in terms of a **goal** users should reach
 - b. Make a draft of an assumed - **expected users' journey**
2. Do **research**: 2 – 6 **interviews** (optimally combined with **observations**) per persona - to explore **real** users' journey: find out what users do, what devices and touchpoints they use, explore users' motivation, painful moments, explore what users think and feel and identify all *insights* and *opportunities*
3. Organize a team **cross-functional workshop** in order to:
 - a. **Visualize** the whole process
 - b. Based on research **insights** make reco how to improve it and fine-tune the user journey process

Journey Steps

STEP 1

STEP 2

STEP 3

Doing

What user's doing (action) during the step

Thinking

What user need to do in detail about the step and action

Feeling



Pain Points

User's frustration. Something that need better solution.

Things to Map

A Traveller

People who love travelling around the world. A frequent traveller.

A Buyer/Requester

People who like to buy something from abroad by entrust others.

A Traveller Journey Steps

BEFORE THE TRIP

DURING THE TRIP

AFTER THE TRIP

BEFORE THE TRIP

Doing

Inform Traveling Schedule

Meet The Buyer

Confirm The Order

Thinking

Post info and T&C via traveling app/web or chat app, inform at social gathering

Ask items details: name, detail, price, photo, etc. Via direct chat or friends

Set up the payment. 50% as upfront or no need down payment at all.

Feeling



Pain Points

Buyer isn't responsive to give confirmation, buyer tend to change items

	DURING THE TRIP		AFTER THE TRIP
Doing	Update The Status	Offer Additional Items	Payment & Delivery
Thinking	Notify buyer when they're OTW, inform current price, send items photo	Offer when desired items unavailable, item's price changes, still more space in luggage Offer items that not heavy & too expensive, send item's photo	Traveler notify the buyer that already come back home & ask delivery method, payment, fee.
Feeling	😊	😞	😊
Pain Points		Difficult to get item since the limited time to shopping, Not responsive buyer	



CUSTOMER JOURNEY MAP

Example of an online grocery store

STAGE	AWARENESS	CONSIDERATION	DECISION		DELIVERY & USE			LOYALTY & ADVOCACY	
CUSTOMER ACTIVITIES	Hear from friends, see offline or online ad, read from newspapers	Compare & evaluate alternatives	Add groceries to shopping cart	Make an order	Receive or pick up on order	Contact customer service	Enjoy groceries	Order again and/or order more	Share experience
CUSTOMER GOALS	No goals at this point	Find the best solution to buy food	Find and select products easily, get inspired	Order effortlessly	Receive or pick up an order effortlessly and when needed	Get help if problems appear, request for refund	Have the right and good quality ingredients	Repeat good customer experience	Share feelings, give feedback
TOUCHPOINTS	Word of mouth, traditional media, social media	Word of mouth, website, brick & mortar store, social media		Website, app, order confirmation email	Delivery service, packing, messages (email, SMS, phone call)	Phone, email, chat	Food products, packages, other materials		Word of mouth, social media
EXPERIENCE	 <i>Interested, curious</i>	 <i>Requires effort but excited</i>	 <i>Excited</i>	 <i>"Payment is painful"</i>	 <i>Requires effort, happy when received</i>	 <i>Frustrated</i>	 <i>Satisfied</i>	 <i>"This is easy"</i>	 <i>"I have to share this"</i>
BUSINESS GOAL	Increase awareness and interest	Increase number of website visitors	Increase shopping cart value & conversion rate	Increase online sales and conversion rate	Deliver on time and minimise the delivery window	Increase customer service satisfaction, minimise waiting time	Make products to match expectations	Increase retention rate and order value and/or frequency	Turn customers into advocates, turn negative experiences into positive
KPIs	Number of people reached	New website visitors	Shopping cart value, conversion rate	Online sales, conversion rate	On time delivery rate, average delivery window	Customer service success rate, waiting time	Product reviews	Retention rate, order value and frequency	Customer satisfaction
ORGANISATIONAL ACTIVITIES	Create marketing campaigns and content both offline and online, PR	Create marketing campaigns and content both offline and online	Optimise grocery shopping experience	Optimise online purchase funnel, order handling	Picking & delivery	Organise customer service	Develop products & product range	Target marketing, make re-ordering easy, upselling and/or cross-selling	Manage feedback and social media, develop sharing / inviting possibilities
RESPONSIBLE	Marketing & Communications	Marketing & Communications	Online development, Customer service	Online development, warehouse, logistics	Warehouse, logistics	Customer service	Product development, purchasing	Marketing, online development	Customer service, online development
TECHNOLOGY SYSTEMS	CRM, analytics, programmatic buying platform, social media	CRM, analytics, CMS, marketing automation	CRM, analytics, CMS, ecommerce platform, PIM	CRM, analytics, CMS, ecommerce platform, PIM, inventory system, marketing automation	CRM, analytics, order & delivery system, marketing automation	CRM, analytics, help desk, ticketing system, chat	CRM, analytics, vendor management system, PIM	CRM, analytics, marketing automation, ecommerce platform	CRM, analytics, marketing automation, ecommerce platform, social media analytics

Task Flow

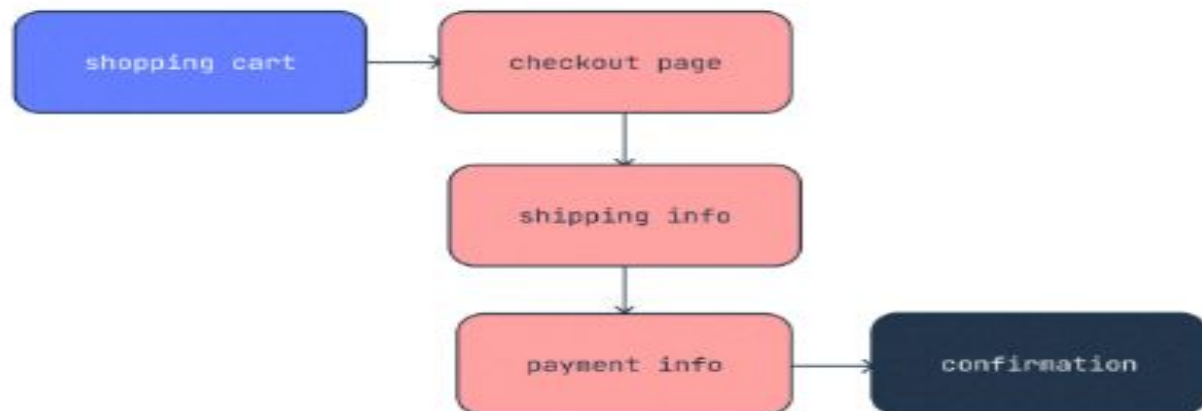
At its core, a task flow is a linear flowchart that outlines the basic path a user takes to complete a specific task within a digital product.

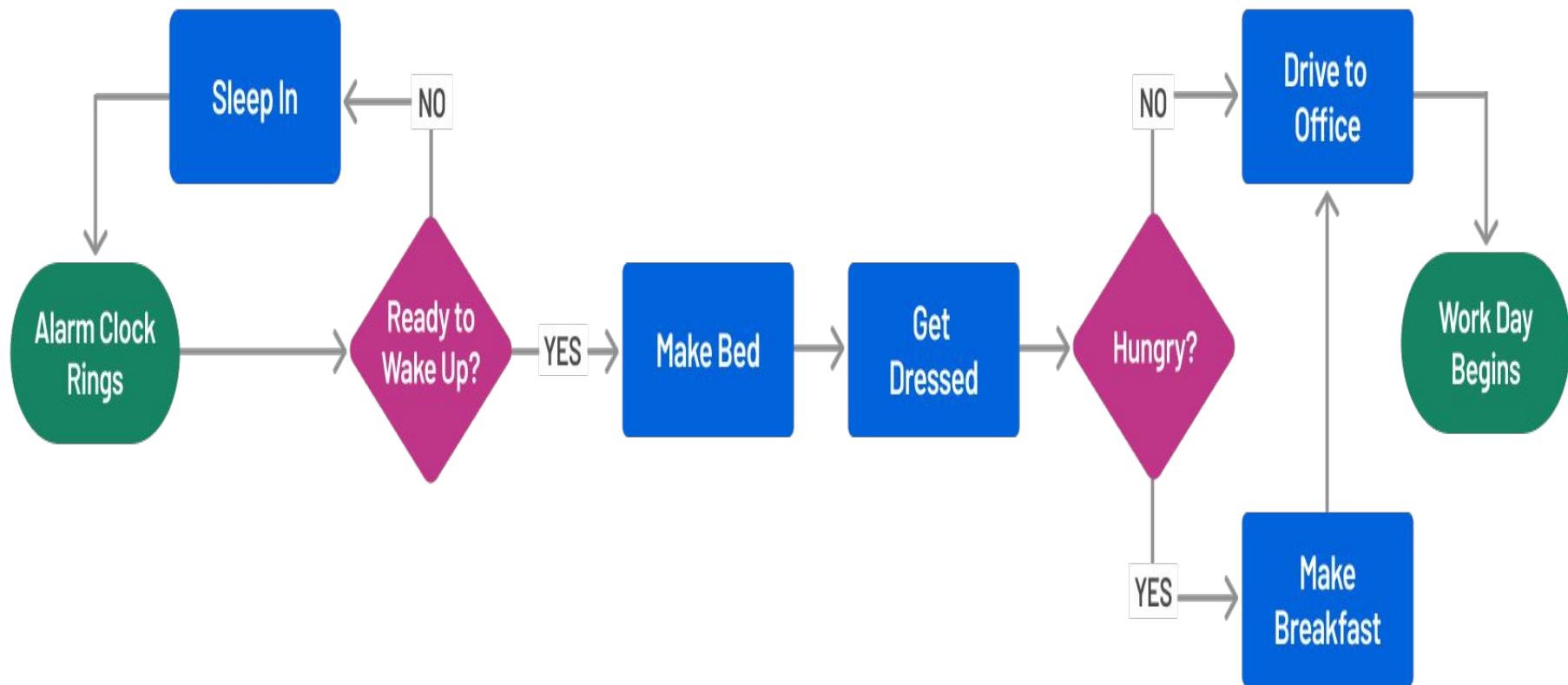
A step-by-step guide, detailing each action they must take to achieve their goal.

Unlike user flows, which focus on include various user choices and multiple paths or a wireflow which gets into UI design, a task flow is a straightforward, high level visual representation of only the essential steps a user takes to reach their goal.

What is task flow?

Task flow



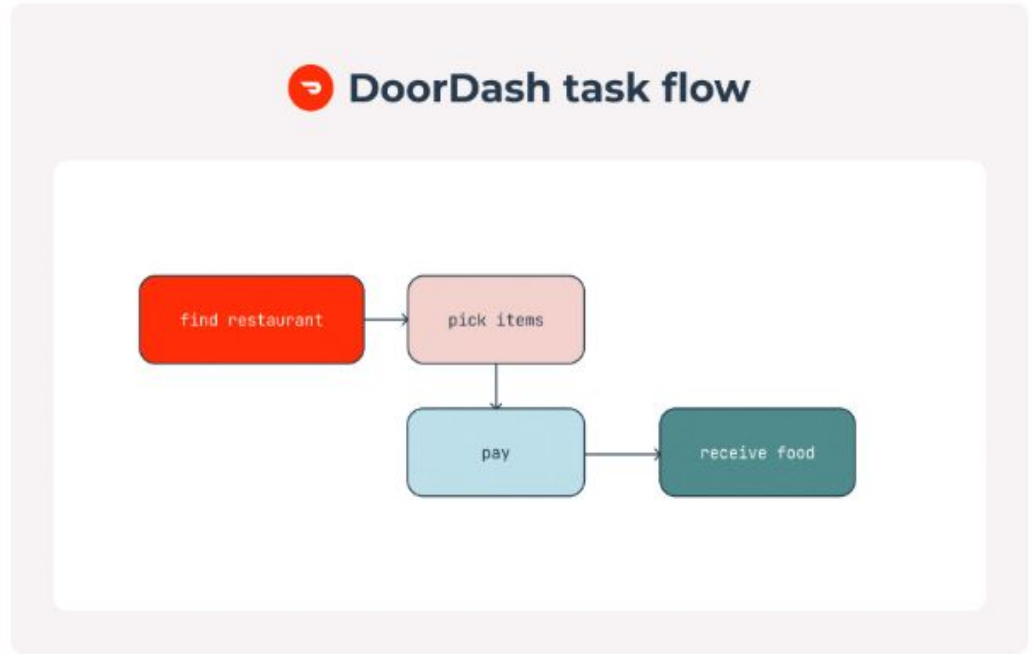


What's the purpose of a task flow?

A task flow diagram aims to give a simplified view of the path a user takes, enabling them to achieve their goals efficiently. Their linear nature, which intentionally exclude design elements and the specifics of user flows, helps in identifying potential pain points, allowing designers to address issues before they become significant problems.

Think of them as a bird's-eye view of task completion, just the highest level steps required to achieve a goal.

If your placing an order on DoorDash, for example, the task flow would be:



Why task flows matter for UX design

Benefits of task flows



Simplify navigation



Boost engagement



Reduce frustration

How to create a task flow

By now you're well aware that a task flow is all about visualizing a particular goal linearly. No multiple paths, no emotional consideration, no fluff – just cold, hard steps.

How to create a task flow



1. Define user goals



2. List actions sequentially



3. Visualize the flow



4. Test and refine



Pain Points Ux

What is a pain point?

Customer pain points are the issues customers experience when using your product or interacting with your service. These problems negatively impact the customer experience and must be addressed to attract new business and improve customer retention.

The 4 types of customer pain points

1



Process

2



Financial

3



Support

4



Product

Types of UX pain points for build brand capability

Following slide showcases different categories of ux pain points to resolve specific customer problems and build brand credibility. It includes challenges such as process, financial, support, productivity etc.

Process pain points



Financial pain points



Support pain points



Productivity pain points



Support pain points-

- o Delayed response for user problems
- o Insufficient product knowledge to users
- o Customer preferred channel not accessible

Financial pain points-

- o Spending excessive amount on membership fees or subscription services
- o Expensive repurchase rates

Productivity pain points-

- o Inconvenient experience while using product
- o Redundant information and resistance to purchase by users

Process pain points-

- o Problem in submitting any applications
- o Difficulty in contacting the appropriate department

Process pain points

Process pain points refer to your organization's internal procedures that hinder the customer journey. The buying process is a frequent contributor to this type of issue and can be indicated by:

Complicated sales transactions

Prolonged deployment times

Disorganized help center

Financial pain points

Financial pain points are associated with the costs of doing business with your company. Your product or service can be exactly what the customer is looking for, but if the price is outside their budget, they'll go to another brand. Financial problems can take the form of:

- ✔ Hidden fees that get added at checkout
- ✔ Expensive membership fees
- ✔ Products that must be frequently replaced

Support pain points

Support pain points refer to the customer's interactions with your sales and customer service teams. Support and process pain points are similar, but support pain points focus on shortcomings in your team's performance rather than company practices. Common support issues are:

- Slow response times
- Poor success rates at resolving issues
- Insufficient product knowledge

Product pain points

Product pain points are any deficiencies in your product or service. They hurt your customer's productivity and prevent them from achieving their desired results. Product problems include:

- ✓ Inconvenient workflows
- ✓ Underwhelming features
- ✓ Inconsistent computer servers

How to identify customer pain points

After learning the types of customer pain points, study your business to learn which problems may be present. There are three general ways to accomplish this.

3 ways to identify pain points



- 1 Gather customer feedback
- 2 Talk to sales and support teams
- 3 Analyze customer data

1. Gather customer feedback

One of the best ways to identify customer pain points is by going directly to your customers. Leverage qualitative market research to learn about the issues customers experience, their current level of satisfaction, and what they might desire moving forward.

Customer satisfaction surveys are a great way to execute this, but rather than asking multiple-choice questions, lean into open-ended questions. For example:

- “What was the greatest struggle during the sales process?”
- “What would you change about our product?”
- “How can our customer service team provide you with a better experience?”

By letting customers answer in their own words, you won't be leading them to make a conclusion you're already aware of. Instead, they're allowed to identify problems you didn't know existed.

Apply the same practice to focus groups and community forums. You can provide topics of conversation and let your customers build on each other's responses to fully understand their perspectives.

2. Talk to sales and support teams

Customers can identify the problems they're experiencing, but they might not always be able to articulate how to fix them. Your sales and customer support teams, on the other hand, have the expertise to provide actionable feedback. They can explain what's going on behind the scenes that may cause inefficiencies.

When you speak with your teams about the customer problems they've seen, apply a similar approach to your open-ended questions as you did with customers. For example:

- ✓ "What problems do you run into most frequently?"
- ✓ "What do customers complain about the most?"
- ✓ "Which changes would you recommend prioritizing?"

3. Analyze customer data

Aside from conducting qualitative research, you'll also want to spend time collecting hard data as well as identifying insights. For example, customer churn rate will tell you how many customers are churning, but it may be more insightful to understand customer churn (and contraction) reasons. This will help fill in the blanks and back up your customers' and employees' claims.

Examining your key performance indicators (KPIs) is a good place to start to see which areas of your operation might be lacking. If you aren't already doing so, consider tracking metrics like:

- Customer churn rate and churn insights
- Average resolution time and resolution time insights
- Conversion rate and conversion insights
- Cart abandonment rate and insights

Many platform analytics can work in real-time to capture these results for you, decreasing the labor required to gather and measure them.

How to address customer pain points

Identifying your pain points isn't the end—it's the beginning. Now you must resolve them and work hard to ensure they don't return.

How to optimize the customer journey



Make it easy
to get help



Use the right
software



Look for pain
point trends



Follow up with
complaints



Send feedback to
the right teams

1. Make it easy to get help

Most customers don't want their hands held every step of the way, but they do want to be attended to when they need it. Self-service features (like knowledge bases) reinforce customer independence and facilitate their search for answers. [Live chat](#) provides a direct channel to support teams.

2. Use the right software

If all your team has is a hammer, they'll treat every problem like a nail. Pain points are unique, so they require unique solutions. Fortunately, there are many software solutions built around solving specific issues:

- ✔ [Call center software](#)
- ✔ [Customer feedback software](#)
- ✔ [Chatbot software](#)
- ✔ [Social media customer service](#)

3. Look for pain point trends

[Measure the customer experience](#) by tracking key metrics to learn where your business might lag. If you identify multiple pain points around a certain process or within a single department, it's a sign that there's a larger issue that needs attention. The responsible team might require:

- Deeper training
- More team members
- Better resources to execute their tasks

4. Follow up with customer complaints

After taking steps to resolve business pain points, talk to your customers. They'll tell you if your actions solved the problem, if it persisted, or if it led to new issues. Keeping the customer's perspective top of mind will reinforce a customer-obsessed culture within your organization.

5. Send feedback to the right teams

Let departments know how the customer perceives their service. Even if a certain pain point might take a long time to fix, there might be things they can do to compensate in the meantime. Likewise, if you receive feedback that the situation is improving, passing that information along can reinforce their behavior.

6. Continuously optimize the customer journey

Keeping your business competitive requires constant attention. Don't neglect areas after you've evaluated them and made improvements—constantly look for ways to deliver better customer experiences. A customer experience strategy can ensure team members stay aligned on company goals and share the responsibility of seeing them through.

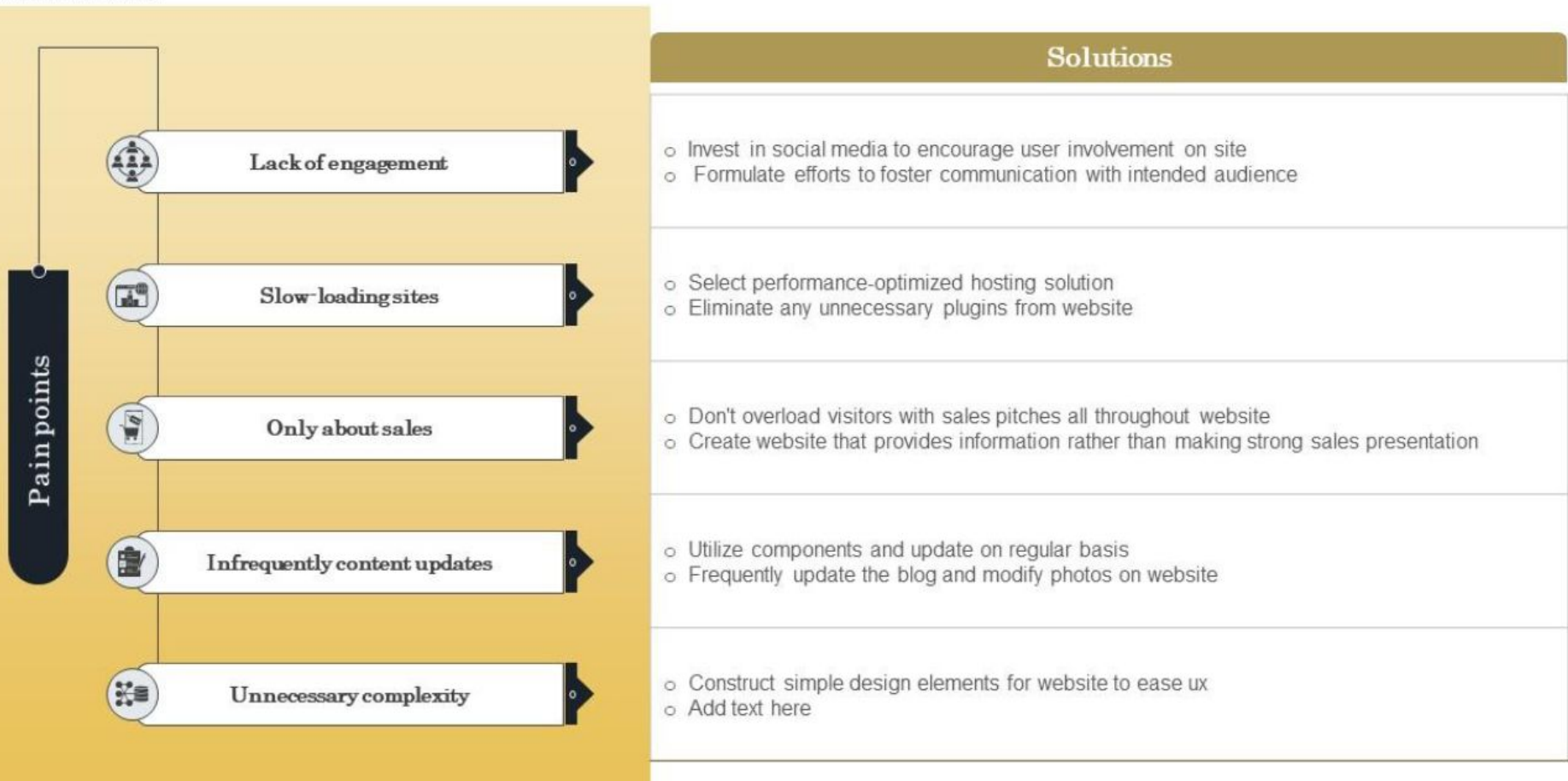
Ux pain points for mobile users with solutions

This slide shows solutions to various issues that mobile user faces to improve the user experience. It includes poor app speed, poor navigation, difficult authentication processes, excessive categorization etc.

Pain points	Key description	Solutions
 Poor app speed	- o Clear wastage of time on page loading due to lagging app - o User move to other app due to poor speed	- o Develop seamless process - o Optimize network speeds and application response times - o Avoid app crashes at all costs
 Poor navigation	- o Significantly decrease in conversion rates - o Poor navigation impacts brand perceptions	- o Simple and intuitive process - o No useless buttons and colours - o Include navigation in structure of app
 Difficult authentication processes	- o Various authentication hurdles due to lengthy procedure - o Drive conversion rates down and compromise security	- o App should not request too many documents - o Simplify authentication as much as possible - o Completely protect against credential attacks
 Excessive Categorization	- o Due to multiple sub categories users find to search anything time consuming - o Add text here	- o Use no excess categorization - o Add text here - o Add text here

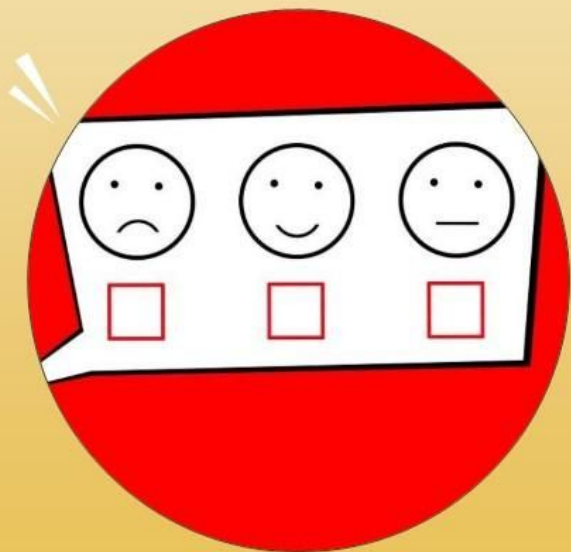
Solutions to overcome pain point for improving ux

Mentioned slide shows multiple issues faced by a customer and solutions to solve the problem. It includes issues such as lack of engagement, infrequent content updates, only about sales, slow loading sites and unnecessary complexity



Methods to identify ux pain points affecting sales and marketing strategy

Mentioned slide depicts various methods that assist business to determine sales and marketing ux pain points. It includes conducting market research, feedback from customers, consult sales and customer service staff and using live chat.



Market research

- Create personas for various customers
 - Most popular
 - High spender
- Conduct qualitative market study
- Map out customer journey to understand ux pain points



Feedback from customers

- Identify and solve problem by listening customer perspective
- Design smooth service interactions to improve ux
- Add text here



Use live chat

- Enable live chat to engage immediately with customers
- Build confidence and trust before providing answers to consumers
- Add text here



Consult sales and customer service staff

- Frequent interact with personnel to gain insights about customer perception
- Add text here
- Add text here

Ux pain points strategies to boost sales

Following slide showcases various strategies to assist business in improving the experience of ux. It comprises of strategies such as website speed, consistency, suggestions, simple and consistent language and ensure ease of use

